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Data Access: Extending Corporate Databases to the Palm Computing® Platform

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Overview

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What questions will be answered?

- What are the benefits?
- What technologies are used; what products are available?
- How do you develop Palm™ applications that make the data useful?
- How is Palm-based data synchronized with server-based data?

Benefits

- Increased productivity:
  - Palm devices are extremely portable
  - Broader deployment options
  - Data where and when you need it
- Lower cost of ownership:
  - Lower cost of devices
  - Lower maintenance costs
  - Improved data integrity
  - See Session 546: “Total Cost of Ownership”
- Scalability:
  - Easier to deploy as demand increases

Enterprise Data

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- Enterprise applications typically store data in a Relational Database (RDBMS):
  - Oracle®, Sybase®, DB2®, etc.
  - Two- or three-tier architectures
- Open standards can be used to access data:
  - SQL
  - ODBC
- Ideally, we need:
  - RDBMS features on the Palm
  - “Simple” data synchronization
  - Database security model on Palm

Solution Types

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- “Extensions” of major RDBMS solutions:
  - Sybase® UltraLite
  - Oracle® 8i Lite
  - IBM® DB2® Everywhere
- 3rd party “forms” based applications with database connectivity:
  - Satellite Forms
  - Pendragon Forms

Data Synchronization

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- Standard HotSync® technology works, but is not great:
  - Requires user's "base" PC to be turned on
  - Single user only
- "Server Sync" solutions offer several benefits:
  - Central administration
  - No requirement for "base" PC
  - Multi-user operation
- Several Server Sync solutions available:
  - Solutions bundled with database solutions
  - 3rd party products available
  - *See Session 528: How to Sync with the Enterprise*

Sybase® SQL Anywhere™ UltraLite

Overview

- Deployment option for SQL Anywhere Studio
- Small footprint relational database optimized for specific application
  - Approximately 50K in footprint
- Use standard SQL, ODBC for building applications
- Synchronizes data with heterogeneous back-end servers
  - Oracle, Microsoft, IBM, Sybase
  - SAP

Sybase® SQL Anywhere™ UltraLite

Building Applications

- Develop application in “C” with CodeWarrior, MobileBuilder
  - Use embedded SQL or ODBC to define queries
- Generate application-optimized database
  - Analyzer determines required features and tables
  - Builds in-memory database
- Link with library created by UltraLite Analyzer

Sybase® SQL Anywhere™ UltraLite

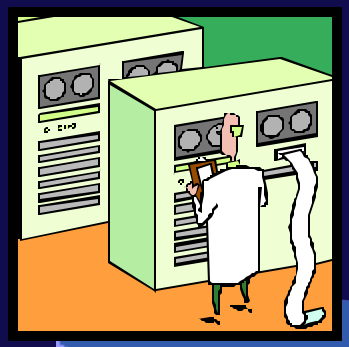
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Data Synchronization

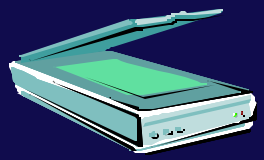
- Synchronizes Palm data with enterprise data sources
  - ODBC
  - ERP
- Connects to Palm device via HotSync conduit or TCP/IP, wireless
- Only changed data synchronized to optimize connection costs
- Conflict detection and resolution

Sybase® SQL Anywhere™ UltraLite

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Enterprise Database / ODBC Data Source

MobiLink Synchronization Server

"SQL" interface

Sybase UltraLite DB

Oracle® 8*i* Lite

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Overview

- Oracle8i Lite is a complete set of libraries for developing, deploying and managing mobile applications on the Palm that synchronize data with Oracle data servers
- Two development options:
  - Oracle8i Lite DBMS for Palm, a 50kb full database for Palm, with indexes and full transactional support
  - Native Palm file formats—develop applications using standard forms based applications
- Oracle8i Lite iConnect:
  - Synchronize data from multiple Palm devices using OKAPI or native Palm file formats (.pdb)

Oracle® 8*i* Lite

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Building Applications

- Oracle8i Lite DBMS (50kb full database for Palm)
  - Applications built using Oracle Kernel API (OKAPI)
  - OKAPI consistent across all Oracle8i Lite platforms
  - Suggest ‘C’ based development environment, i.e., CodeWarrior
- Using native Palm file format
  - Use any development environment to create applications
  - Oracle8i Lite iConnect will replicate such data to Oracle data servers

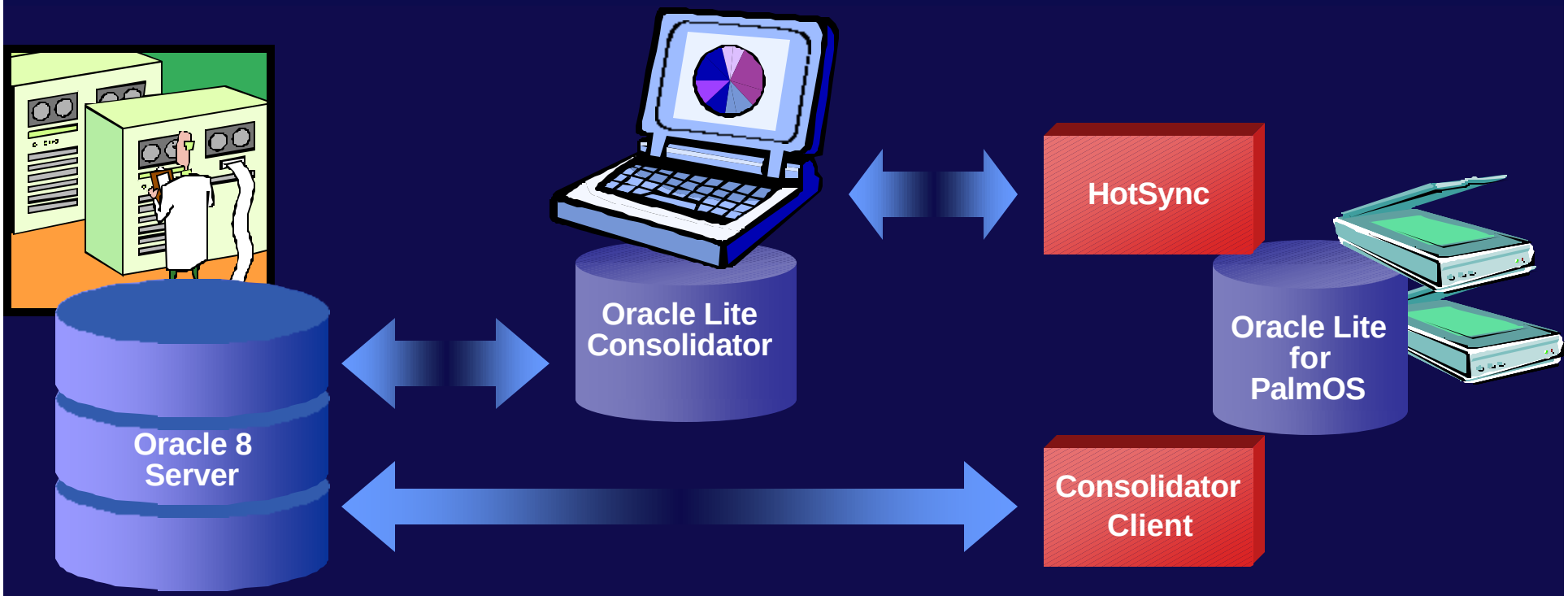
Oracle® 8*i* Lite

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Data Synchronization

- Simple HotSync or network HotSync to Oracle8i Lite client
- Network TCP/IP Sync
  - Consolidator Client implemented on Palm as separate application
  - Synchronizes data from Oracle Lite databases, native Palm OS databases, Satellite Forms data
- Supports third party conduits if required by customer, i.e., Riverbed Scout
- Centralized management and deployment

Oracle® 8*i* Lite



IBM® DB2® Everywhere

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Overview

- Small footprint relational database for the Palm Computing® platform
- Server-based sync using Mobile Connect™
  - DB2
  - Lotus Notes
  - MS Exchange
  - ODBC
- Applications developed with Satellite Forms or with “C”, “C++” (CodeWarrior)
- QBE application available for Palm OS

IBM® DB2® Everywhere

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Building Applications

- Develop applications in “C” or “C++” using CodeWarrior
  - Call Level Interface
  - Subset of ODBC
- Develop forms-based applications using Satellite Forms
  - Available 12/99

IBM® DB2® Everywhere

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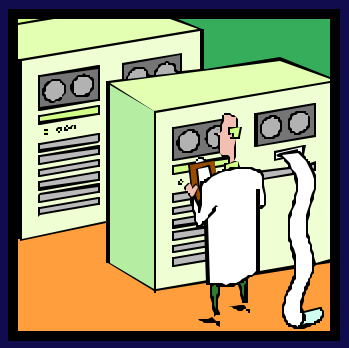
Data Synchronization

- Mobile Connect performs two-way, server based synchronization between Palm device and:
  - DB2 for OS/390
  - DB2 for AS/400
  - DB2 Universal for UNIX, OS/2, Windows NT
  - Any ODBC-compliant data source
- Conflict resolution rules defined with Mobile Connect

IBM® DB2® Everywhere

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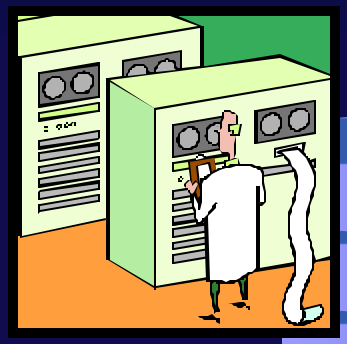


Satellite Forms

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- A Form-based application development tool with a runtime environment for Palm OS™
- Table schemas can be imported using ODBC to help define forms
- Conduit can be built using ActiveX to synchronize with dBase5
- dBase5 can be linked to other RDBMS system
- Server-based synchronization available with Intellisync™ Anywhere

Satellite Forms



Enterprise Database

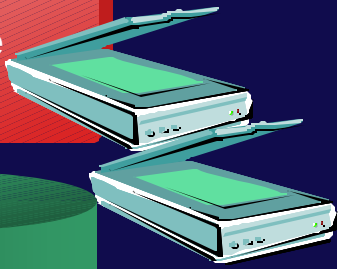
ActiveX API

dBase5 Database

Conduit

Form Definitions

Runtime Engine



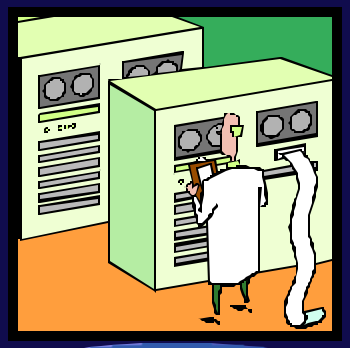
PalmOS Database

Pendragon Forms

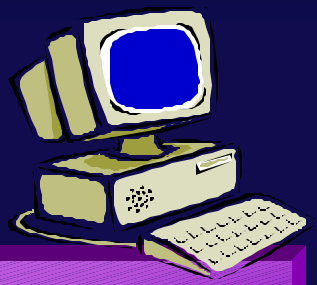
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- A Form-based application development tool with a runtime environment for Palm OS
- Table schemas can be imported from any ODBC data source to help define forms
- Conduit synchronizes Palm data with any OBDC data source and/or Access
- Integrated with the WaveSync™ server from WaveWare™ for server synchronization

Pendragon Forms



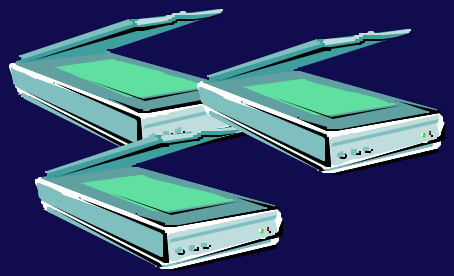
ODBC Data Source



WaveSync Server

Pendragon Conduit

Via Remote Access Service





HotSync

ndragon Conduit

Via Direct Connection



Review

- Solutions from database vendors include data storage and synchronization:
  - Sybase SQL Anywhere UltraLite
  - Oracle Consolidator, Oracle Lite
  - IBM DB2 Anywhere
- Forms-based packages provide integration to ODBC data sources:
  - Satellite Forms with ActiveX API
  - Pendragon Forms with ODBC conduit


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